

University of Kent

Individual Report

Airbnb

Viola Romoli

Social Media Marketing and Web Analytics

Due date: 11/03/2026

Table of Contents

1. Introduction and Analytical Objective	3
2. Data and Methodology	5
3. Analysis and Findings	6
3.1 Sentiment Analysis	6
3.2 Topic Modelling	7
3.2.1 Seasonal Topic Modelling.....	10
3.3 Regression Analysis	12
4. Interpretation and Strategic Insight	12
5. Recommendations	13
6. Limitations and Reflection	14
7. Conclusion	15
References.....	15

Analysis of Airbnb's Guest Reviews to Improve Their Social Media Strategy

1. Introduction and Analytical Objective

Airbnb is an American company that operates as an online marketplace for short- and long-term homestays, experiences and services in various countries and regions. This model has completely changed the market for tourism accommodation because it allows ordinary people to rent out their houses, apartments, or rooms as accommodations for tourists at a lower price compared to hotels ('peer-to-peer accommodation'), focusing also more on the interaction between guest and host. Previously, this model had to face some limitations, such as the difficulty of the host in making their accommodations known to potential guests, and the challenge of establishing trust between the host and guests. Airbnb overcame these obstacles thanks to the internet technologies, such as the company website and social media platforms, which provide visibility (Guttentag, 2015).

Moreover, the online reviews that users can leave on Airbnb's website and social media platforms (especially Instagram and TikTok) play a central role in the tourism and hospitality industry, because travellers increasingly rely on online content when they want to search where to stay and what to do during their trips. In fact, according to Xiang and Gretzel (2010), social media and online review platforms have become key sources of travel information, and they influence consumers' evaluation of destinations, services, and experiences.

Nowadays, social media and guest reviews are also important for both the company and hosts, because social media allows the company to reach a wider range of audience and guest reviews are important for hosts to understand customer satisfaction because they provide a valuable insight about customer experiences and perceptions. According to Hootsuite (2026), Airbnb social media platforms, especially Instagram, perform well because people talk positively about the brand, and they are engaged with the content they publish, meaning that they constantly attract new users (see Figure 1). However, comparing Airbnb's performance with two of its competitors, Vrbo and Booking, it is possible to see that during a 30-day period, Booking dominates in terms of online presence, despite the 9.3% decline over the period, while Airbnb led in engagement with 6.8M interactions and 117.3% growth rate, showing a stronger audience interaction. Overall, all three platforms produce a positive sentiment and good performances, but Vrbo remain a marginal presence across all metrics (see Figure 2).

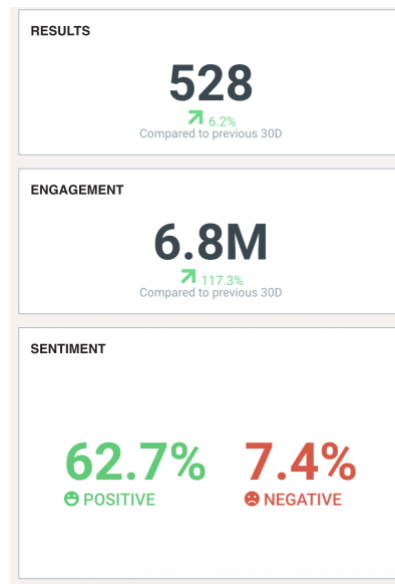


Figure 1 – Airbnb's 30-day Instagram key metrics on Hootsuite

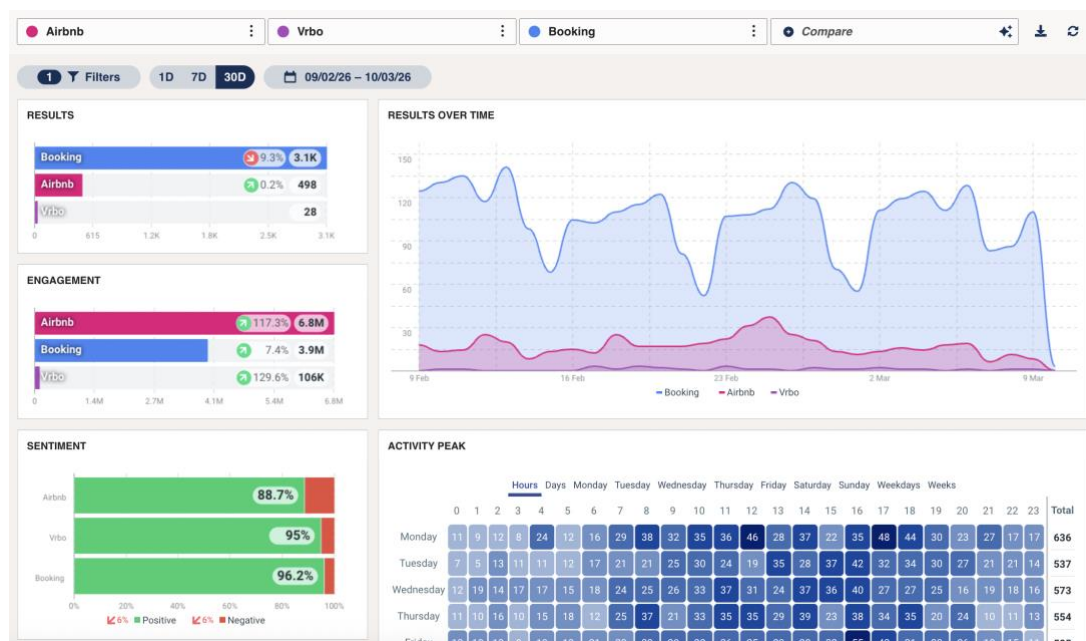


Figure 2 – Vrbo and Booking's Instagram key metrics on Hootsuite

However, even if Airbnb's strategy works, the content that it posts on its social media is primarily focused on showing experiences and houses that it is possible to rent around the world (see Figure 3). Thus, this report, through an analysis of Airbnb's guest reviews, aims to answer a specific analytics problem: **which themes in Airbnb guest reviews are most strongly associated with positive sentiment?** The identification of these themes aims to provide strategic insight for creating social media content that might attract even more users and improve the platform's engagement.

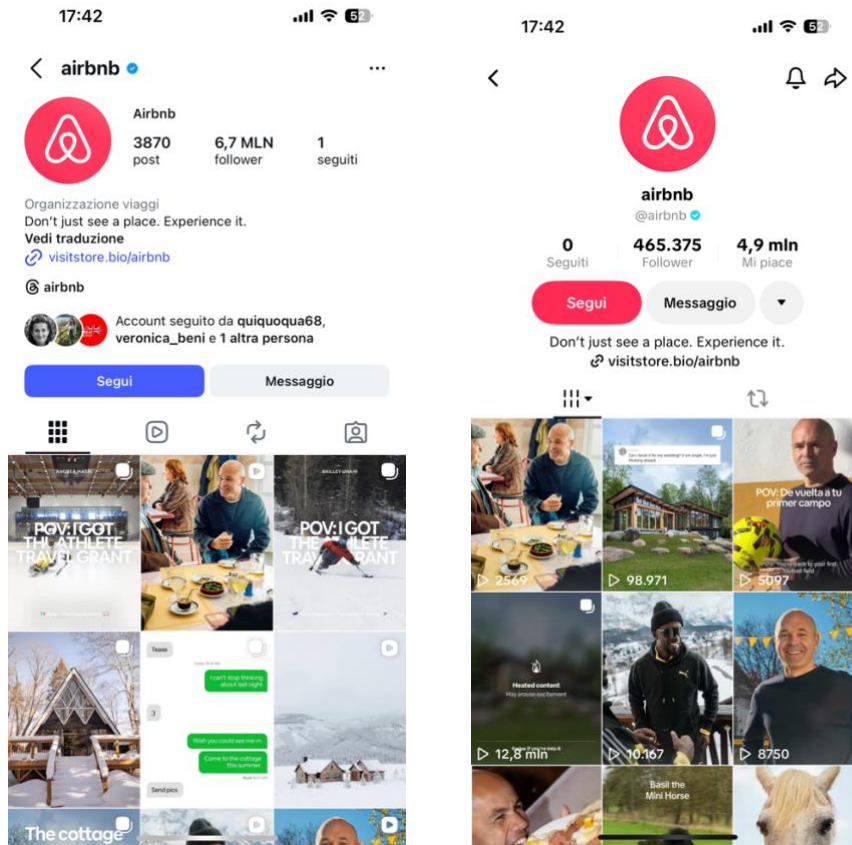


Figure 3 – Airbnb's Instagram and TikTok feeds

2. Data and Methodology

To conduct this analysis, data from the website 'Inside Airbnb' has been used. It was decided to use the dataset 'reviews.csv' for the city of London, which contains detailed data about the reviews written by the customers (e.g. date, comments, reviewer_name, etc.). The review dataset contained 2097996 comments written between 21/12/2009 and 17/09/2025, but it was decided to reduce the period from 17/09/2024 to 17/09/2025 to focus on more up-to-date reviews; consequently, the number of reviews was reduced to 550795. The reviews have also been filtered by language using Google's Compact Language Detector (library cld2) to eliminate non-English reviews because the R packages used for the sentiment analysis are based exclusively on English dictionaries. Thus, a word in another language would be scored 0, altering the average. Then, the data were cleaned of punctuation, numbers, non-English characters, multiple spaces, URLs, emails, and empty comments. Finally, comments have been converted into lowercase.

For the actual analysis, a combination of multiple analytical techniques has been used to examine the reviews in the dataset and find the themes which are associated with positive sentiment. First, sentiment analysis has been applied to find the average sentiment score, which represents the overall emotional tone of the comment, assigning a numerical score that

reflects whether guests expressed positive (greater than 0), negative (below 0), or neutral (close to 0 or equal to 0) opinions about their stay. Then, a topic modelling has been used to find the most recurrent topic among the reviews, showing which themes guests most frequently discuss. Two other specific analyses have been done: one to identify the predominant topics across every season and one to find if customers write more about physical structures or human interactions. The second one has been carried out by classifying reviews with the highest number of relevant words representing the two categories. Finally, a linear regression analysis has been employed to quantify the link between review topics and sentiment (Sponder and Khan, 2018).

The analyses have been conducted using the programming language R because it allows to analyse in a flexible and personalised way datasets that are the result of the datafication process, which is the transformation of social action into online quantified data (Van Dijck, 2014).

3. Analysis and Findings

3.1 Sentiment Analysis

The results of the sentiment analysis reveal an overall positive perception among the guests (see Figure 4). The mean sentiment score of 0.69 and the median of 0.67 are aligned, and they suggest a relatively symmetric distribution skewed towards positive values. The first quartile also has a score of 0.42, which indicates that the bottom 25% of the reviews also express a slightly positive sentiment. The minimum score of -1.91 shows the presence of some negative reviews, but the number is low because the score of the first quartile is already positive. The third quartile captures most of the reviews, 75% are below 0.95 of the sentiment score; joining this information with the mean and median score, this highlights a generally satisfying guest experience. Finally, the maximum score of 3.56 suggests that a reduced group of guests expressed particularly enthusiastic feedback.

It was also possible to analyse the sentiment score over the time of the year taken into consideration, and it revealed a stability in the average scores, with values between 0.68 and 0.71 from September 2024 to August 2025. There is also a slight dip during the winter season and a peak in September (see Figure 5).

```
> summary(airbnb_reviews$sentiment_score)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
-1.9125  0.4243  0.6740  0.6934  0.9508  3.5567
```

Figure 4 – Summary of sentiment analysis

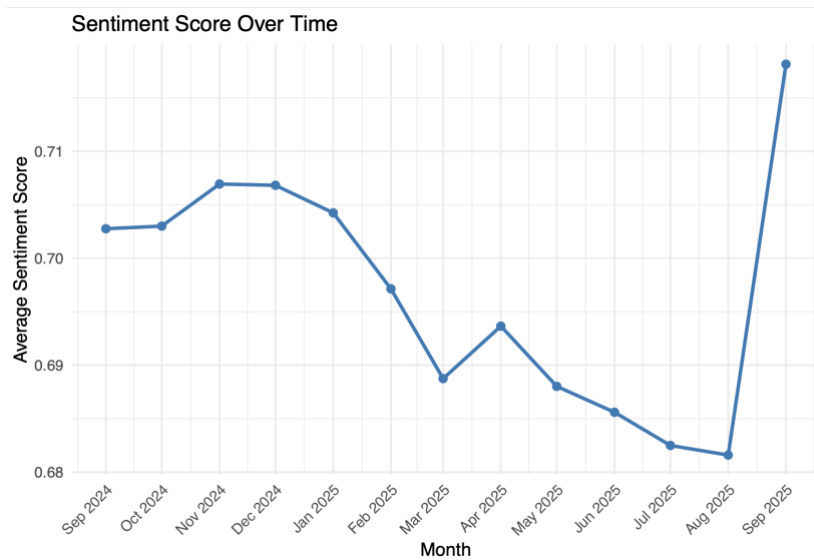


Figure 5 – Sentiment score over time

3.2 Topic Modelling

The results of the topic modelling identified eight different topics across the Airbnb reviews, revealing what customers discuss about their stay in London (see Figures 6 and 7). The most relevant topic (Topic 6) focuses on location and connectivity with words such as *location*, *walk*, *station* and *central*, showing how proximity to transport links and main services (e.g. restaurants) is what customers care more about. Topic 7 is also related to transportation and accessibility because it presents words such as *good*, *room*, *station* and *easy*. Topic 8 and Topic 4 indicate an overall positive experience with terms such as *love*, *stay*, *clean*, *comfort*, and *recommend*, suggesting that most guests were satisfied with the structure and the stay because they were clean and pleasant. Topic 3 also expresses enthusiasm for the experience, but with a focus on the neighbourhood where the customers stayed, with words such as *nearby* and *street*. Topic 2 highlights terms such as *responsive*, *kind*, and *communication*, showing that customers care also about the relationship with the host. Topic 5 focuses more on logistical and practical aspects of the stay because it includes references to *bed*, *kitchen*, *area*, and *arrival*, highlighting the importance of practicality during the stay. In the end, topic 1 stands out as the most critical because it presents words such as *small*, *issue* and *noise*, indicating that practical concerns and minor inconveniences can lead to a slight dissatisfaction. To contextualise the content of all eight topics, it was necessary to read some examples of comments. Also, it was analysed how topics change over time, and it is possible to notice that almost all topics are stable over time; there are no drastic changes in the review's content. The only one that shows some fluctuations is Topic 2, which increases during the summer but decreases again during winter (see Figure 8).

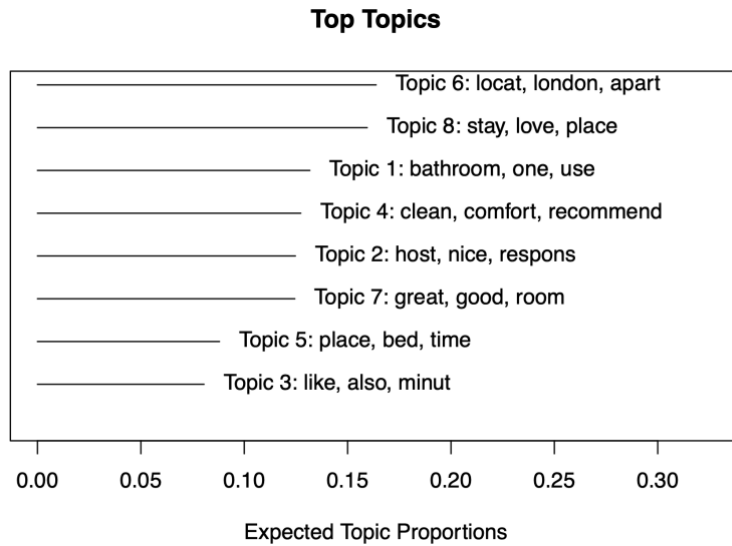


Figure 6 – Shows how much space each topic occupies in the set of reviews

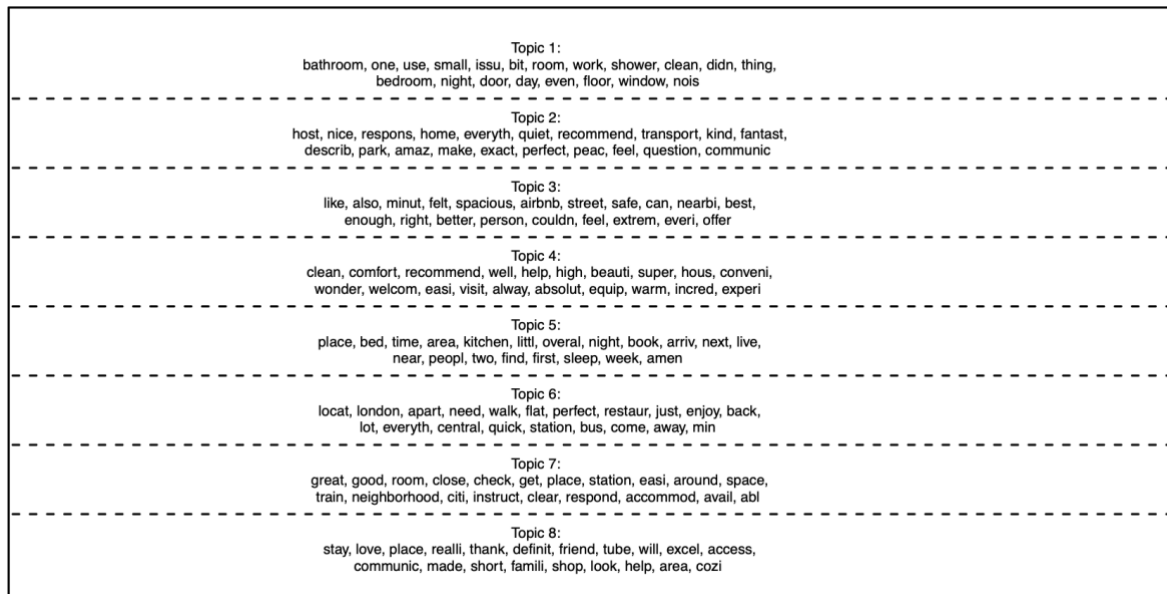


Figure 7 – Shows the 20 most representative words per topic

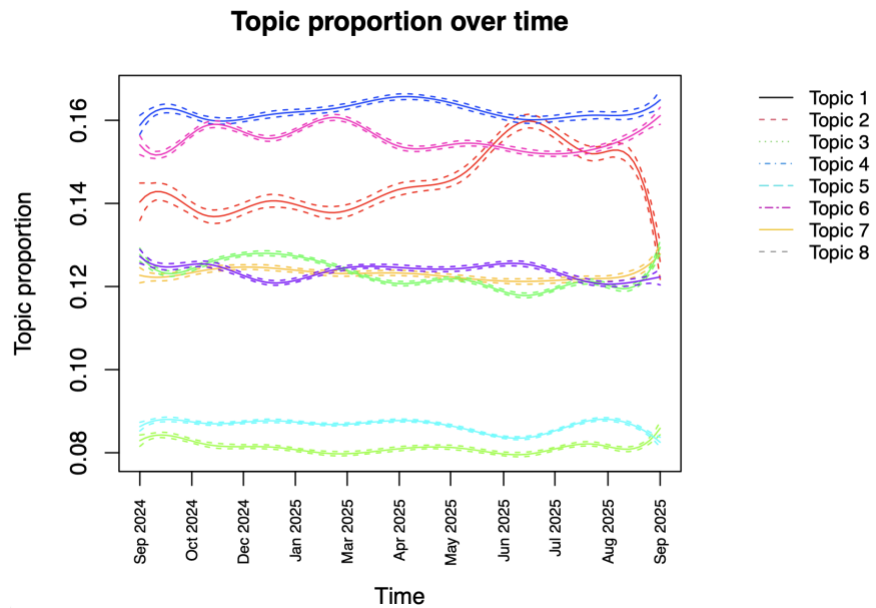


Figure 8 – Topic proportion over time

Since the analysis wants to identify which topics are more associated with positive sentiment, the relationship between sentiment and topics has been analysed. As it is possible to see in Figure 9, Topics 4, 2, and 8 are the ones most associated with a positive sentiment, while Topic 1 is the most strongly associated with negative sentiment. Topics 6, 3 and 7 have a middle position, indicating that these topics are discussed regardless of sentiment, both in negative and positive reviews.

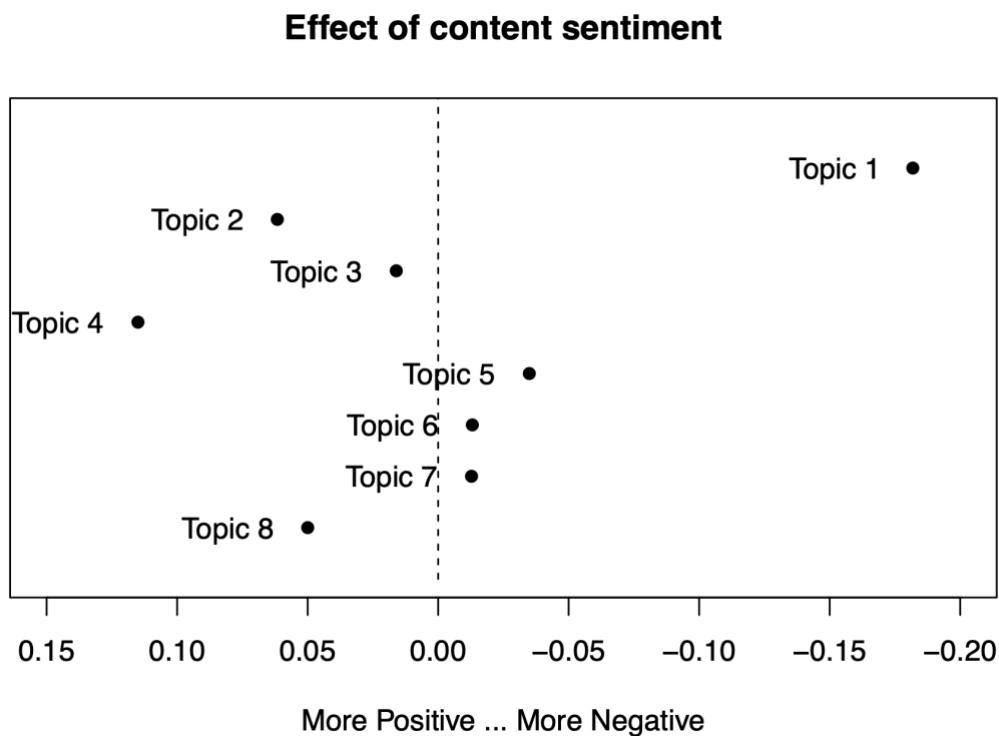


Figure 9 – Effect of sentiment on topic prevalence

3.2.1 Seasonal Topic Modelling

An additional analysis has been conducted to identify how topics change according to the seasons and which ones are more related to positive sentiment. The results of the analysis also show that across all seasons, most of the topics are associated with positive sentiment, and only one is strongly associated with a negative sentiment (see Figure 10). Through the analysis of the representative words of all topics and sample comments, it is possible to notice that across seasons, the negative sentiment is predominantly related to technical problems with the structure, while positive sentiment might vary slightly across seasons. Some of the most positive topic are related with a general satisfaction during the stay and the quality and comfort of the structure; but, there are also more seasonal related topics, such as topics related to location and neighbourhood during summer and spring (Topic 2 in spring and Topic 5 in summer), meaning that some part of the city are more alive or aesthetically beautiful during certain period of the year, or topic related with hospitality during winter (Topic 2), indicating that guests may spent more time in the structure due to the weather, increasing the possibility to interact with the host.

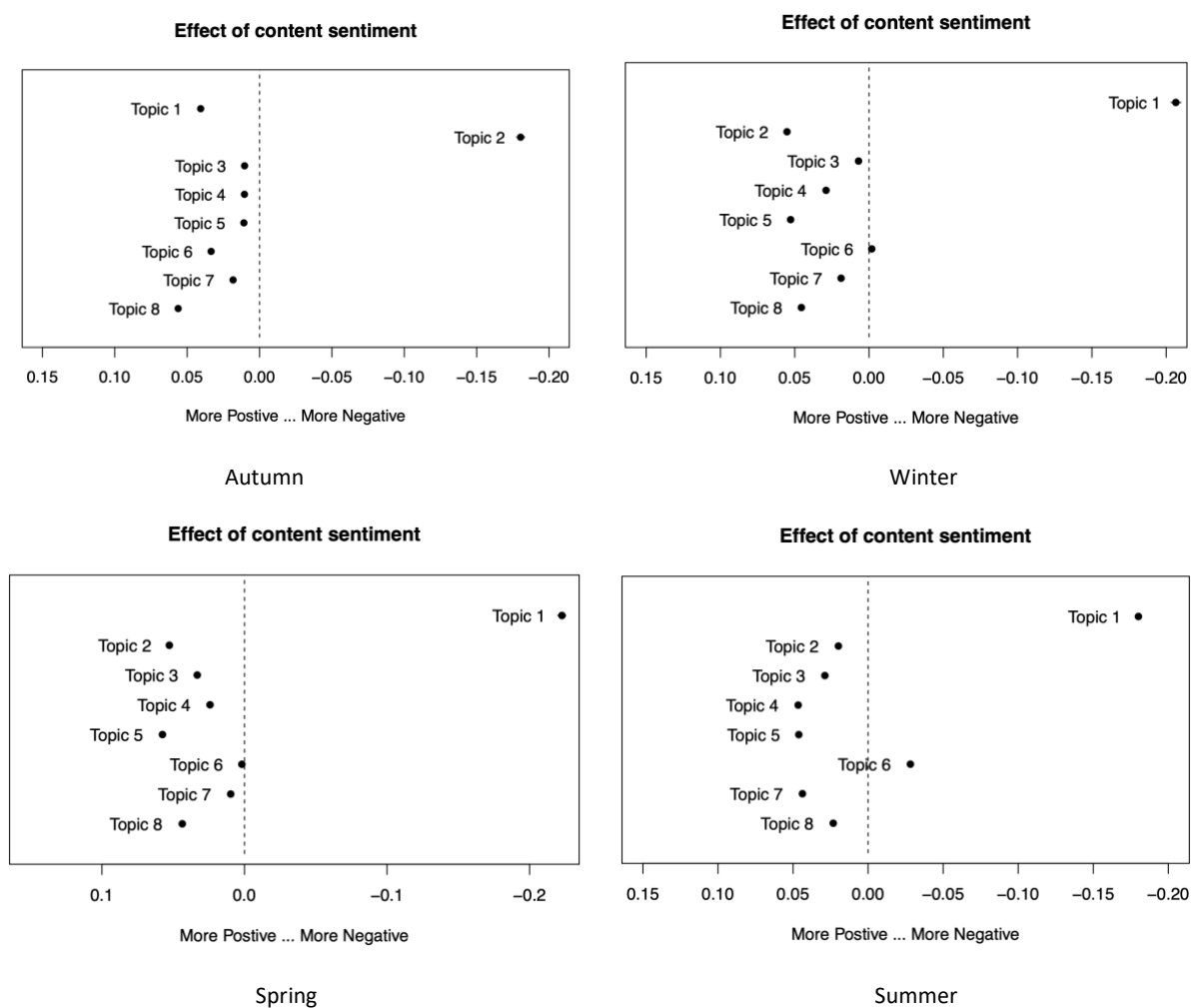


Figure 10 – Effect of sentiment on topic prevalence in Autumn, Winter, Spring and Summer



Figure 11 – Shows the 20 most representative words in Autumn, Winter, Spring and Summer

Analysing in-depth the words and samples of the reviews, it is also possible to see how topics vary and acquire different facets based on the season; for this reason, it is complex to compare them in detail. Considering the two key recurrent aspects in Airbnb reviews, it is possible to determine if across the different seasons customers emphasise more the host or the structure and which one generates a higher positive sentiment. The results show that customers mention the physical structure slightly more than the interaction with the host. It is also interesting to notice that reviews usually focus on both aspects instead of one of the two, and this is expressed by the balanced label (see Figure 12).

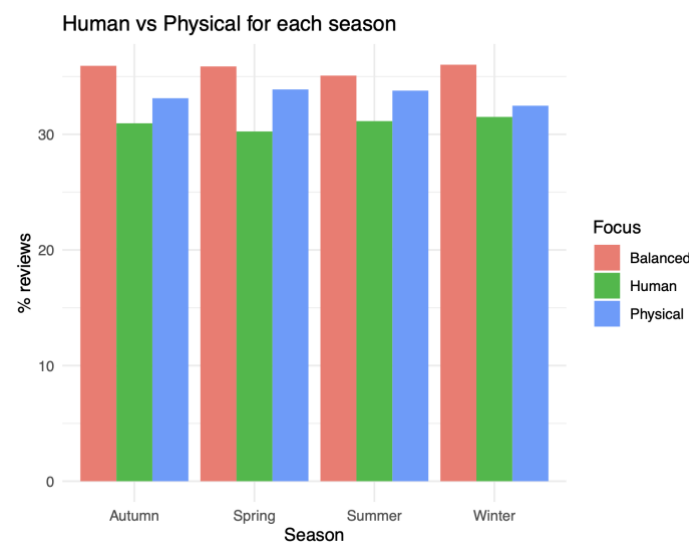


Figure 12 – Human vs physical parameter for each season

3.3 Regression Analysis

The results of the regression analysis conducted on the year (without the seasonal distinction) confirm and quantify the relationship between topic prevalence and guest sentiment, following a linear regression model. They show that Topic 4 is the strongest predictor of positive sentiment, with a coefficient of 10.05, indicating that cleanliness and comfort are what satisfy guests the most. Topic 3 and Topic 7 also show positive coefficients of 5.44 and 5.16, respectively, suggesting that those who talk about the surrounding area tend to express a higher level of satisfaction. Topic 6 and Topic 2 contribute more moderately, with coefficients of 1.67 and 1.08, which indicate that location and relationship with the host are important but have less impact in shaping the overall tone of the review. Topic 5 is the only topic with a negative coefficient of -0.32, which indicates that when customers talk about functional and logistical aspects of the stay, their sentiment is slightly lower, probably because these elements are mentioned more frequently when expectations have not been fully met (e.g. problems with check-in). Finally, Topic 1 has a moderately positive coefficient, but it is the topic with the most negative sentiment score; this reflects the fact that guests sometimes mention concerns in positive reviews (see Figure 13).

Coefficients: (1 not defined because of singularities)					
	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-2.36902	0.01523	-155.53	<2e-16	***
Topic1	2.42394	0.01520	159.51	<2e-16	***
Topic2	1.08306	0.02962	36.56	<2e-16	***
Topic3	5.44806	0.01790	304.44	<2e-16	***
Topic4	10.05871	0.01968	511.09	<2e-16	***
Topic5	-0.32407	0.02660	-12.19	<2e-16	***
Topic6	1.67581	0.02208	75.89	<2e-16	***
Topic7	5.16348	0.02282	226.25	<2e-16	***
Topic8	NA	NA	NA	NA	

Figure 13 – Regression analysis coefficients

4. Interpretation and Strategic Insight

The combination of the findings from the sentiment analysis, topic modelling, and regression analysis shows what Airbnb guests in London are more interested in, providing useful information to improve their social media channels.

The overall positive sentiment, represented by 75% of reviews below 0.95 of sentiment score and the mean and median score, suggests that London Airbnb Properties meet guests' expectations. Airbnb reviews are also a valuable form of earned media, which is content generated by guests on which neither host nor platforms have direct control (Chaffey and Ellis-

Chadwick, 2022). These reviews show the genuine guest satisfaction that can be quantified with the sentiment score calculated through the sentiment analysis.

The topic modelling and regression analysis show that not all positive experiences have the same weight in shaping the overall satisfaction. Through social listening, which is the analysis of user-generated content to find strategic themes rather than simply monitoring individual mentions (Tuten, 2024), the eight topics that have been identified show key aspects of a stay. Cleanliness and comfort (Topic 4) emerge as the principal topic related to positive sentiment, with a regression coefficient of 10.05, nearly double that of the next-highest predictors. This finding indicates that the host needs to invest in property maintenance and meticulous cleaning protocols because they are what drive enthusiastic reviews. Other topics that contribute positively are related to the neighbourhood and local transport (Topics 3, 6, and 7), thus hosts need to strongly emphasise them in their advertisements. Host communication (Topic 2) is a particularly interesting case, because while its regression coefficient is moderate (1.08), it is among the top topics associated with positive sentiment and shows seasonal fluctuation with a peak in summer when there are more guests. Thus, maintaining a good communication quality could lead to a return in positive review sentiment. Topic 5, which is related to functional and logistical aspects, has a slightly negative coefficient (-0.32), suggesting that these elements emerge most prominently when something goes wrong. Simplifying check-in procedures and ensuring that all things in the house function well would reduce the possibility of these topics appearing in a negative context.

The seasonal focus shows that negative sentiment across all seasons is related to technical problems and property maintenance; thus, if hosts manage to fix these problems, this could meaningfully reduce the minority of negative reviews that affect the overall score. Moreover, the seasonally related topics in winter, summer, and spring are an opportunity to differentiate social media content during the year. In the end, the balance between host-related and structure-related comments underlines that guests evaluate both dimensions simultaneously. Thus, the best thing to do is to integrate both: hosts have to combine a clean and well-equipped property with a friendly and responsive communication, especially during peak seasons, to consistently generate the enthusiastic, high sentiment reviews that Topic 4 and Topic 2 represent.

5. Recommendations

Based on the analytical findings, it is possible to provide some practical recommendations for Airbnb's social media channels (Instagram and TikTok), taking into consideration that according to Kietzmann et al. (2011), the honeycomb framework different platforms emphasise different social behaviour and for this they require different content strategies.

First, Airbnb should focus more, both on Instagram and TikTok, on emphasising a cleanliness and comfort aesthetic of the houses since they represent the main topic associated with

positive guest sentiment. Thus, on Instagram, where visual identity and reputation are the dominant aspects, they might increase the type of content where they show clean interiors, arranged amenities, and comfortable spaces such as a bed with fresh linen and well-lit spaces, while on TikTok, where sharing and entertainment lead to engagement, they can create short videos of “property walkthrough” or “before and after cleaning” to show their commitment to the topic. They already do this kind of content, but over a period of three months, they did it only two times; it is not enough if this is the main topic that creates positive sentiment, and that could increase the traffic to Airbnb.

Second, they can use a seasonal narrative to diversify their content during the year because the seasonal analysis shows that there are some slight differences in topic discussion across the seasons. Airbnb should publish content emphasising cool, well-ventilated spaces during summer and content highlighting warmth and cosiness during winter. They could also do some videos related to maintenance to show that the houses are regularly checked during the year, addressing in this way the negative comments about technical problems. This approach aligns with the principle of social listening because it might transform guest concerns into a content strategy that builds trust before guests even arrive (Tuten, 2024).

Third, Airbnb could create content related to the relationship with the host since it is a topic consistently associated with positive sentiment. Both on TikTok and Instagram, they could do videos and stories featuring real hosts welcoming guests, showing their structures, or talking about their city, to reinforce the relational dimension. This type of content might be useful because on social media, users are usually in a discovery mindset; they do not think about purchasing, but engaging, low-friction content could attract their attention and drive them to the profile where they might convert into potential guests.

6. Limitations and Reflection

The analysis presented in this report is subject to some methodological and data limitations. Hootsuite was used as a third-party tool to see Airbnb’s social performance. As a third-party tool, Hootsuite can only give a general idea about social media performance because it proposes aggregated data that hides specific trends, and its metrics are pre-defined according to the platform’s scope rather than the researcher’s analytical needs. Moreover, the free trial version shows only data from the last 30 days, and TikTok data were entirely unavailable, representing a significant gap. Due to these limitations, R was used instead because it permits a more flexible, personalised, and complex analysis. This had been achieved by running the scripts provided in class, but they have been adapted to the specific analytical problem that has been investigated.

The dataset used for the analysis, retrieved from the ‘Inside Airbnb’ website, also presented some limitations because a UK dataset was unavailable; thus, the London dataset has been used as an example, but this may limit the generalisation of the findings to a broader market.

A future analysis would benefit from collecting data from different countries to see if topics and interests profoundly vary among them. If so, a good social media strategy to take advantage of this could be to create different personalised pages for every country that can focus on specific interests.

7. Conclusion

In conclusion, this report identified which themes in Airbnb guest reviews are most strongly associated with positive sentiment, to use this insight to improve Airbnb's social media content strategies. This has been done through sentiment analysis, topic modelling, and regression analysis of Airbnb reviews. The results of the analysis showed that in the guest reviews prevail a positive sentiment and that this sentiment is mainly related to cleanliness, comfort, location, and interaction with the host. Thus, based on the results, Airbnb's social media content on Instagram and TikTok should be more aligned with these themes that guests themselves identify as most satisfying. The seasonal dimension further enriches the strategy, indicating specific types of content to increase engagement with users. In the end, this analysis also demonstrates the importance of applying data-driven social listening to earned media because guest reviews become a source of strategic insight that can help improve the strategy of social media content. In this way, Airbnb can align its communication with what guests are really interested in.

References

- Chaffey, D. and Ellis-Chadwick, F. (2022). *Digital Marketing: Strategy, Implementation and Practice*. 8th ed. Harlow: Pearson.
- Guttentag, D. (2015). Airbnb: Disruptive Innovation and the Rise of an Informal Tourism Accommodation Sector. *Current Issues in Tourism*, 18(12), pp.1192–1217.
doi:<https://doi.org/10.1080/13683500.2013.827159>.
- Hootsuite (2026). *Social Media Marketing & Management Dashboard*. [online] Hootsuite. Available at:
<https://www.hootsuite.com/?srsltid=AfmBOoo5wf9tOY78l8ktSshBVu1lcp9ZpYZqW8VM4QoSF7H3DZxVNe2t> [Accessed 7 Mar. 2026].

Kietzmann, J.H., Hermkens, K., McCarthy, I.P. and Silvestre, B.S. (2011). Social media? Get serious! Understanding the Functional Building Blocks of Social Media. *Business Horizons*, 54(3), pp.241–251. doi:<https://doi.org/10.1016/j.bushor.2011.01.005>.

Sponder, M. and Khan, G.F. (2018). *Digital analytics for marketing*. New York; London: Routledge.

Tuten, T.L. (2024). *Social Media Marketing*. 5th ed. London: Sage.

Van Dijck, J. (2014). Datafication, Dataism and Dataveillance: Big Data between Scientific Paradigm and Ideology. *Surveillance & Society*, 12(2), pp.197–208. doi:<https://doi.org/10.24908/ss.v12i2.4776>.

Xiang, Z. and Gretzel, U. (2010). Role of Social Media in Online Travel Information Search. *Tourism Management*, 31(2), pp.179–188. doi:<https://doi.org/10.1016/j.tourman.2009.02.016>.